



Vivekanand Education Society's Institute of Technology
An Autonomous Institute Affiliated to University of Mumbai

**End Semester Examination
Summer 2024**

Max marks: 60

Branch: Computer Engineering

Name of the Course: Operating System

Course code: CSC404

Duration: 2 hours

Semester: IV

QP Code: R23-CSC404_022023-24

- N.B.** (1) Attempt any three out of the five questions.
(2) Figures to the right indicate full marks.
(3) Assume suitable data if necessary.

	Marks
Q.1 (a) Explain the difference between Microkernel, Monolithic and Layered.	05
(b) What is meant by system call? Discuss about types of system calls	05
(c) Consider the following set of process with the length of CPU burst time given in milliseconds: Process(P1,P2,P3,P4,P5), Arrival Time(2,3,0,4,3), Burst Time(2,3,1,2,2) & Priority (3,2,4,1,3). Draw the three Gantt charts illustrating the execution of these Processes using SJF, Preemptive priority, Round Robin (quantum=2) scheduling and calculate waiting and turnaround time of each process for each scheduling algorithm.	10
Q.2 (a) What is a process? Explain about various fields of Process Control Block	05
(b) Explain Multi - Instance Resource Allocation Graph with an example.	05
(c) What is the Dining Philosophers problem? Discuss the solution to the Dining philosopher's problem using Semaphores.	10
Q.3 (a) Given free memory partitions of 100 K, 500 K, 200 K, 300 K, and 600 K (in order), how would each of the First-fit, Best-fit, and Worst-fit algorithms place processes of 212 K, 417 K, 112 K, and 426 K (in order)?	05
(b) Explain the difference between External fragmentation and Internal fragmentation with a suitable diagram.	05
(c) Consider the following page reference string: 1,2,3,4,2,1,5,6,2,1,2,3,7,6,3,2,1,2,3,6. How many page faults would occur for the optimal page replacement algorithm, assuming three frames and all frames are initially empty.	10

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- Q.4 (a) How does Belady's anomaly challenge traditional assumptions about the relationship between the number of page frames and the occurrence of page faults in virtual memory management? Explain with Example 10
- (b) Explain the three allocation methods in file system implementation. Illustrate with proper diagrams. 10
- Q.5 (a) Compare the LOOK and C-LOOK disk scheduling algorithms. How do they differ from their counterparts, SCAN and C-SCAN? 10
- (b) Compare various functions of Special Purpose Operating System 10

X—X—X—X